Kubernetes Basics Bootcamp - Lab Report

Introduction This report documents the completion of the Kubernetes basics tutorial series, including the Hello Minikube guide and Modules 2 through 6 of the Kubernetes Bootcamp. The experiments were conducted on a Windows machine using PowerShell as the shell, with minikube running a local Kubernetes cluster via the Docker driver.

Experiment Steps Summary

Part 1: Hello Minikube The lab began with setting up a minikube cluster using minikube start and verifying its status with minikube status. A Deployment named hello-node was created using kubectl create deployment with the agnhost test image. I verified the Pod status using kubectl get pods, checked application logs with kubectl logs, and exposed the application externally using kubectl expose deployment with --type=LoadBalancer. The application was accessed via minikube service hello-node, which opened a tunnel due to the Docker driver isolation on Windows.

Part 2: Deploying Your First App (Module 2) I created a new Deployment named kubernetes-bootcamp using the image gcr.io/google-samples/kubernetes-bootcamp:v1. I verified the Deployment and Pod status, then opened a second terminal to run kubectl proxy to establish a connection between the local machine and the Kubernetes cluster's internal network. I accessed the Kubernetes API version endpoint via curl http://localhost:8001/version to confirm the proxy was working. I then retrieved the Pod name and accessed the application through the proxy API endpoint.

Part 3: Viewing Pods and Nodes (Module 3) I explored the cluster components by running kubectl get pods, kubectl get nodes, and kubectl describe pods to inspect detailed Pod information including IP addresses, ports, and lifecycle events. I viewed application logs using kubectl logs. I also executed commands inside the container using kubectl exec, first non-interactively to check environment variables, and then interactively using kubectl exec -ti to open a bash shell inside the Pod. Inside the container, I inspected the application source code (server.js) and tested the application locally using curl http://localhost:8080.

Part 4: Using a Service to Expose Your App (Module 4) I created a NodePort Service for the kubernetes-bootcamp Deployment using kubectl expose deployment with --type=NodePort. Due to the Docker driver on Windows, direct access via minikube ip was not possible, so I used minikube service kubernetes-bootcamp --url to create a tunnel and obtain a local URL. I verified the Service was load-balancing traffic by making multiple curl requests. I then explored Kubernetes Labels by examining the automatically created label app=kubernetes-bootcamp on the Deployment, queried Pods and Services using label selectors (-l app=kubernetes-bootcamp), and applied a custom label version=v1 to a Pod using kubectl label pods. Finally, I deleted the Service using kubectl delete service -l app=kubernetes-bootcamp and confirmed that external access was no longer possible while the application continued running internally.

Part 5: Running Multiple Instances (Module 5) I scaled the kubernetes-bootcamp Deployment from 1 to 4 replicas using kubectl scale deployments/kubernetes-bootcamp --replicas=4. I observed the ReplicaSet behavior and verified that four Pods were running. I tested the built-in load balancing of the Service by executing multiple curl requests and observing that each request was served by a different Pod instance. I then scaled the Deployment down to 2 replicas and confirmed that two Pods were gracefully terminated while two remained running.

Part 6: Rolling Update (Module 6) I performed a rolling update by changing the container image from v1 to v2 using kubectl set image. I monitored the rollout progress with kubectl rollout status deployments/kubernetes-bootcamp and observed that Kubernetes incrementally replaced old Pods with new ones without downtime. I verified the update by checking the Pod images and curling the application to confirm the version changed to v2. Next, I simulated a failed update by attempting to deploy a non-existent image tag (v10). As expected, the new Pods entered an ImagePullBackOff state. I then rolled back to the previous stable version using kubectl rollout undo deployments/kubernetes-bootcamp, which restored the Deployment to v2. Finally, I cleaned up all resources using kubectl delete deployments/kubernetes-bootcamp services/kubernetes-bootcamp.

Problems Encountered and Reflections

Problem 1: Incorrect Pod Name in kubectl logs When attempting to view logs with kubectl logs, I encountered a "NotFound" error because I mistyped the Pod name. Specifically, I confused similar-looking characters such as 4dfc versus 64fc and 1th6z versus lth6z. The solution was to always copy the exact Pod name from the output of kubectl get pods rather than typing it manually. I also learned to use Tab auto-completion in PowerShell to avoid such typos.

Problem 2: PowerShell Syntax Differences from Linux Shell The tutorial commands were written for POSIX shells (bash), but I was using PowerShell. Commands using export, $(), angle brackets <>, and certain quoting conventions failed. For example, kubectl logs <pod-name> failed because PowerShell interprets <> as redirection operators. The solution was to remove the angle brackets entirely and pass the Pod name directly. For environment variables, I replaced export VAR=value with PowerShell's $env:VAR = value syntax.

Problem 3: kubectl proxy Not Running When trying to access http://localhost:8001/version, I received a connection error because kubectl proxy was not running in a separate terminal. The tutorial explicitly requires a second terminal for the proxy, but it is easy to overlook. The solution was to open a new PowerShell window, run kubectl proxy, leave it running, and then execute curl commands in the original terminal.

Problem 4: go-template and jsonpath Quoting Issues in PowerShell

Several commands using -o go-template or -o jsonpath failed with parsing errors because PowerShell handles single quotes and escape sequences differently from bash. For instance, template strings containing \n or " caused unrecognized character errors. The solution was to simplify the queries. Instead of complex go-templates, I used -o jsonpath='{.items[0].metadata.name}' where possible, or alternatively, I ran kubectl get pods manually and copied the name directly when the automated variable assignment proved unreliable.

Problem 5: Docker Driver Network Isolation on Windows When using curl http://$(minikube ip):$NODE_PORT, the connection failed because minikube runs inside a Docker container on Windows, and the host machine cannot directly access the internal Docker network IP. This is a known limitation of the Docker driver. The solution was to use minikube service <service-name> --url in a separate terminal to create a tunnel, which provides a 127.0.0.1 URL accessible from the host. This approach was necessary every time external access was required.

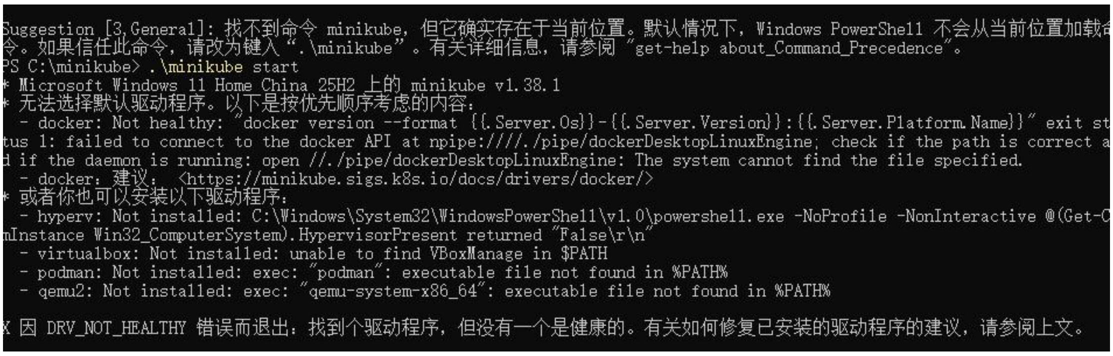
Problem 6: curl Not Found Inside Container When executing kubectl exec to run curl inside the container for internal connectivity testing, the command failed with "executable file not found in $PATH". The solution was to recognize that lightweight container images do not always include curl. As an alternative, I verified application availability by checking Pod logs with kubectl logs, or by entering the container shell with kubectl exec -ti and using available tools like wget or Node.js built-in HTTP capabilities.

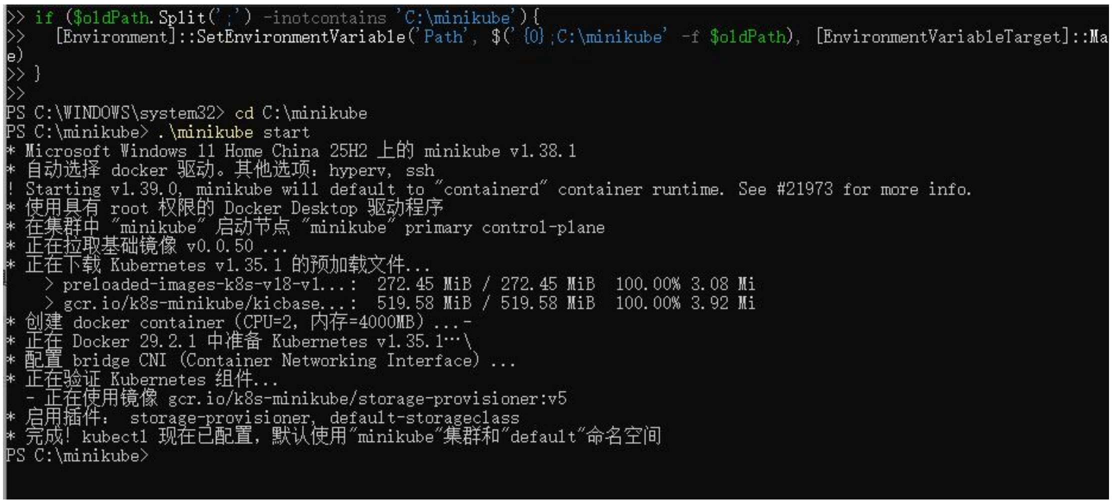
Problem 7: minikube service Blocking the Terminal I accidentally ran minikube service in the main terminal, which blocked the command prompt because the tunnel process must remain active. The solution was to press Ctrl+C to stop it, then either run it in a dedicated secondary terminal or capture the URL from the output and use it directly in the main terminal for subsequent curl commands.

Problem 8: Accessing the Wrong Pod via Proxy When querying the Pod through the proxy API, I initially accessed the wrong Pod (balanced from a previous experiment) instead of kubernetes-bootcamp. This happened because the environment variable $env:POD_NAME contained the wrong Pod name. The solution was to explicitly filter for the correct Pod name when setting the variable, ensuring I targeted the intended application.

Conclusion This bootcamp provided hands-on experience with core Kubernetes concepts including Deployments, Pods, Services, Labels, scaling, and rolling updates. Working through these exercises on Windows with PowerShell highlighted the importance of understanding shell-specific syntax differences and driver-specific networking behaviors. The troubleshooting process reinforced the value of carefully reading command outputs, using kubectl get and kubectl describe for diagnostics, and understanding that Kubernetes abstractions like Services and Deployments provide resilience and flexibility for application management.

Screenshot：





```
PS C:\minikube> kubectl get po -A
NAMESPACE     NAME                               READY   STATUS    RESTARTS   AGE
kube-system   coredns-7d764666f9-lwrzh           1/1     Running   0          113s
kube-system   etcd-minikube                      1/1     Running   0          2m
kube-system   kube-apiserver-minikube            1/1     Running   0          2m
kube-system   kube-controller-manager-minikube   1/1     Running   0          2m
kube-system   kube-proxy-g6z4d                   1/1     Running   0          113s
kube-system   kube-scheduler-minikube            1/1     Running   0          2m
kube-system   storage-provisioner                1/1     Running   0          117s
PS C:\minikube>
```


```
PS C:\minikube> kubectl get po -A
NAMESPACE     NAME                               READY   STATUS    RESTARTS   AGE
kube-system   coredns-7d764666f9-lwrzh           1/1     Running   0          113s
kube-system   etcd-minikube                      1/1     Running   0          2m
kube-system   kube-apiserver-minikube            1/1     Running   0          2m
kube-system   kube-controller-manager-minikube   1/1     Running   0          2m
kube-system   kube-proxy-g6z4d                   1/1     Running   0          113s
kube-system   kube-scheduler-minikube            1/1     Running   0          2m
kube-system   storage-provisioner                1/1     Running   0          117s
```


```
PS C:\minikube> kubectl create deployment hello-minikube --image=kicbase/echo-server:1.0
deployment.apps/hello-minikube created
PS C:\minikube>
```

```
Request served by hello-minikube-58f7c595dd-77d7p

HTTP/1.1 GET /

Host: 127.0.0.1:50097
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate, br, zstd
Accept-Language: zh-CN,zh;q=0.9,en;q=0.8,en-GB;q=0.7,en-US;q=0.6
Connection: keep-alive
Sec-Ch-Ua: "Chromium";v="148", "Microsoft Edge";v="148", "Not/A)Brand";v="99"
Sec-Ch-Ua-Mobile: ?0
Sec-Ch-Ua-Platform: "Windows"
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: none
Sec-Fetch-User: ?1
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36 Edg/148.0.0.0
```

```
PS C:\WINDOWS\system32> kubectl create deployment balanced --image=kicbase/echo-server:1.0
>> kubectl expose deployment balanced --type=LoadBalancer --port=8080
>>
deployment.apps/balanced created
service/balanced exposed
PS C:\WINDOWS\system32> minikube tunnel
>>
* 隧道成功启动

* 注意：请不要关闭此终端，因为此进程必须保持活动状态才能访问隧道......

* 为服务 balanced 启动隧道。
```

```
PS C:\WINDOWS\system32> kubectl get services balanced
>>
NAME       TYPE           CLUSTER-IP    EXTERNAL-IP   PORT(S)          AGE
balanced   LoadBalancer   10.99.81.79   127.0.0.1     8080:31750/TCP   58s
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> minikube addons enable ingress
>>
* ingress 是由 Kubernetes 维护的插件。如有任何问题，请在 GitHub 上联系 minikube。
您可以在以下链接查看 minikube 的维护者列表：https://github.com/kubernetes/minikube/blob/master/OWNERS
* 插件启用后，请运行 "minikube tunnel" 您的 ingress 资源将在 "127.0.0.1"
  - 正在使用镜像 registry.k8s.io/ingress-nginx/controller:v1.14.3
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
* 正在验证 ingress 插件...
```

```
NAME       TYPE           CLUSTER-IP     EXTERNAL-IP    PORT(S)          AGE
balanced   LoadBalancer   10.99.81.79    127.0.0.1      8080:31750/TCP   58s
PS C:\WINDOWS\system32> minikube addons enable ingress
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  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
* 正在验证 ingress 插件...
* 启动 'ingress' 插件
```

```
PS C:\WINDOWS\system32> minikube addons enable ingress
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  - 正在使用镜像 registry.k8s.io/ingress-nginx/controller:v1.14.3
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
* 正在验证 ingress 插件...
* 启动 'ingress' 插件
PS C:\WINDOWS\system32> kubectl apply -f https://storage.googleapis.com/minikube-site-examples/ingress-example.yaml
pod/foo-app created
service/foo-service created
pod/bar-app created
service/bar-service created
ingress.networking.k8s.io/example-ingress created
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> minikube addons enable ingress
* ingress 是由 Kubernetes 维护的插件。如有任何问题，请在 GitHub 上联系 minikube。
您可以在以下链接查看 minikube 的维护者列表：https://github.com/kubernetes/minikube/blob/master/OWNERS
* 插件启用后，请运行 "minikube tunnel" 您的 ingress 资源将在 "127.0.0.1"
  - 正在使用镜像 registry.k8s.io/ingress-nginx/controller:v1.14.3
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
* 正在验证 ingress 插件...
* 启动 'ingress' 插件
PS C:\WINDOWS\system32> kubectl apply -f https://storage.googleapis.com/minikube-site-examples/ingress-example.yaml
pod/foo-app unchanged
service/foo-service unchanged
pod/bar-app unchanged
service/bar-service unchanged
ingress.networking.k8s.io/example-ingress unchanged
PS C:\WINDOWS\system32> kubectl wait --for=condition=ready --timeout=300s ingress/example-ingress
```

```
PS C:\WINDOWS\system32> minikube addons enable ingress
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您可以在以下链接查看 minikube 的维护者列表：https://github.com/kubernetes/minikube/blob/master/OWNERS
* 插件启用后，请运行 "minikube tunnel" 您的 ingress 资源将在 "127.0.0.1"
  - 正在使用镜像 registry.k8s.io/ingress-nginx/controller:v1.14.3
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
* 正在验证 ingress 插件...
* 启动 'ingress' 插件
PS C:\WINDOWS\system32> kubectl apply -f https://storage.googleapis.com/minikube-site-examples/ingress-example.yaml
pod/foo-app unchanged
service/foo-service unchanged
pod/bar-app unchanged
service/bar-service unchanged
ingress.networking.k8s.io/example-ingress unchanged
PS C:\WINDOWS\system32> kubectl wait --for=condition=ready --timeout=300s ingress/example-ingress
PS C:\WINDOWS\system32> kubectl get ingress
NAME              CLASS   HOSTS   ADDRESS        PORTS   AGE
example-ingress   nginx   *       192.168.49.2   80      4m17s
PS C:\WINDOWS\system32> kubectl get pods -n ingress-nginx
NAME                                        READY   STATUS      RESTARTS   AGE
ingress-nginx-admission-create-wrmr5        0/1     Completed   0          8m6s
ingress-nginx-admission-patch-69b64         0/1     Completed   2          8m6s
ingress-nginx-controller-596f8778bc-brhdp   1/1     Running     0          8m6s
```

```
PS C:\WINDOWS\system32> minikube tunnel
* 隧道成功启动

* 注意：请不要关闭此终端，因为此进程必须保持活动状态才能访问隧道......

* 为服务 balanced 启动隧道。
! 在 Windows 上使用 v8.1以上版本的OpenSSH客户端，访问 1024 以下端口可能会失败。更多信息请参阅：https://minikube.sigs.
k8s.io/docs/handbook/accessing/#access-to-ports-1024-on-windows-requires-root-permission
* 为服务 example-ingress 启动隧道。
```

```
Windows PowerShell
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安装最新的 PowerShell，了解新功能和改进！https://aka.ms/PSWindows
                        minikube version
>> C:\WINDOWS\system32>
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
PS C:\WINDOWS\system32> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\WINDOWS\system32> _
```

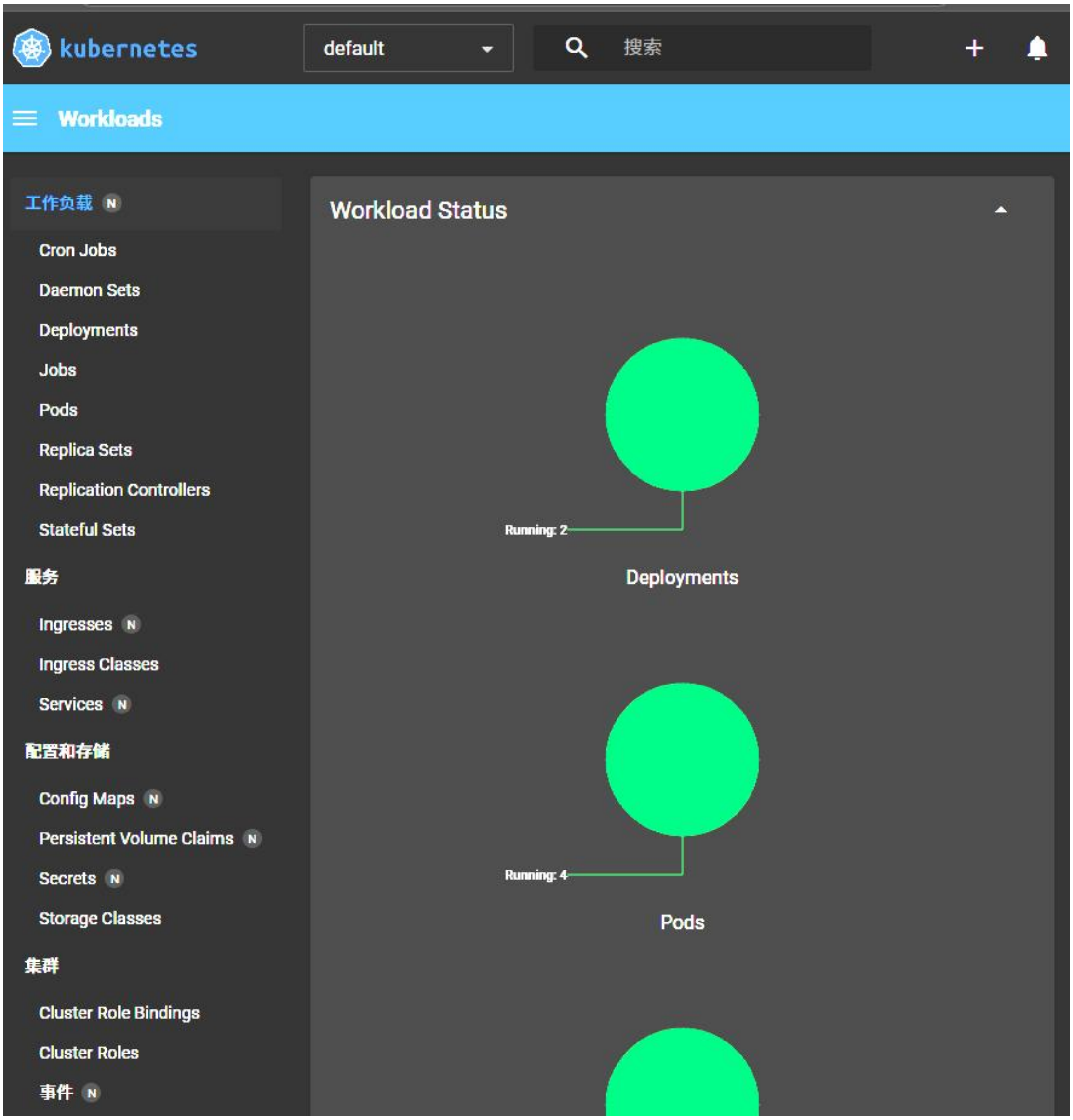
```
                        minikube version
>> C:\WINDOWS\system32>
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
PS C:\WINDOWS\system32> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\WINDOWS\system32> minikube start
* Microsoft Windows 11 Home China 25H2 上的 minikube v1.38.1
* 根据现有的配置文件使用 docker 驱动程序
* 在集群中 "minikube" 启动节点 "minikube" primary control-plane
* 正在拉取基础镜像 v0.0.50 ...
* 正在更新运行中的 docker "minikube" container .../
* 正在 Docker 29.2.1 中准备 Kubernetes v1.35.1…-
* 正在验证 Kubernetes 组件...
  - 正在使用镜像 gcr.io/k8s-minikube/storage-provisioner:v5
* 插件启用后，请运行 "minikube tunnel" 您的 ingress 资源将在 "127.0.0.1"
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/controller:v1.14.3
* 正在验证 ingress 插件...
* 启用插件： storage-provisioner, default-storageclass, ingress
* 完成！kubectl 现在已配置，默认使用"minikube"集群和"default"命名空间
PS C:\WINDOWS\system32> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

PS C:\WINDOWS\system32> _
```

```
PS C:\WINDOWS\system32> minikube dashboard
* 正在开启 dashboard ...
  - 正在使用镜像 docker.io/kubernetesui/dashboard:v2.7.0
  - 正在使用镜像 docker.io/kubernetesui/metrics-scraper:v1.0.8
* 某些仪表板功能需要 metrics-server 插件。要启用所有功能，请运行：

        minikube addons enable metrics-server

* 正在验证 dashboard 运行情况 ...
* 正在启动代理...
* 正在验证 proxy 运行状况 ...
_
```




```
PS C:\WINDOWS\system32> minikube dashboard
* 正在开启 dashboard ...
  - 正在使用镜像 docker.io/kubernetesui/dashboard:v2.7.0
  - 正在使用镜像 docker.io/kubernetesui/metrics-scraper:v1.0.8
* 某些仪表板功能需要 metrics-server 插件。要启用所有功能，请运行：

        minikube addons enable metrics-server

* 正在验证 dashboard 运行情况 ...
* 正在启动代理...
* 正在验证 proxy 运行状况 ...
* 正在使用默认浏览器打开 http://127.0.0.1:57284/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashb
oard:/proxy/ ...
```

```
PS C:\WINDOWS\system32> kubectl create deployment hello-node --image=registry.k8s.io/e2e-test-images/agnhost:2.53 --
/agnhost netexec --http-port=8080
deployment.apps/hello-node created
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl create deployment hello-node --image=registry.k8s.io/e2e-test-images/agnhost:2.53 --
/agnhost netexec --http-port=8080
deployment.apps/hello-node created
PS C:\WINDOWS\system32> kubectl get deployments
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
balanced         1/1     1            1           28m
hello-minikube   1/1     1            1           32m
hello-node       1/1     1            1           77s
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> kubectl get pods
NAME                              READY   STATUS    RESTARTS        AGE
balanced-774d97997-mc8zf          1/1     Running   1 (7m50s ago)   29m
bar-app                           1/1     Running   1 (7m50s ago)   24m
foo-app                           1/1     Running   1 (7m50s ago)   24m
hello-minikube-58f7c595dd-77d7p   1/1     Running   1 (7m49s ago)   32m
hello-node-64fc5894d-2ndhh        1/1     Running   0               98s
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> kubectl get events
LAST SEEN   TYPE     REASON                    OBJECT                                MESSAGE
29m         Normal   Scheduled                 pod/balanced-774d97997-mc8zf          Successfully assigned default/
balanced-774d97997-mc8zf to minikube
7m43s       Normal   Pulled                    pod/balanced-774d97997-mc8zf          Container image "kicbase/echo-
server:1.0" already present on machine and can be accessed by the pod
7m42s       Normal   Created                   pod/balanced-774d97997-mc8zf          Container created
7m42s       Normal   Started                   pod/balanced-774d97997-mc8zf          Container started
7m45s       Normal   SandboxChanged            pod/balanced-774d97997-mc8zf          Pod sandbox changed, it will b
e killed and re-created.
29m         Normal   SuccessfulCreate          replicaset/balanced-774d97997         Created pod: balanced-774d9799
7-mc8zf
29m         Normal   ScalingReplicaSet         deployment/balanced                   Scaled up replica set balanced
-774d97997 from 0 to 1
24m         Normal   Scheduled                 pod/bar-app                           Successfully assigned default/
bar-app to minikube
7m35s       Normal   Pulled                    pod/bar-app                           Container image "docker.io/kic
base/echo-server:1.0" already present on machine and can be accessed by the pod
7m35s       Normal   Created                   pod/bar-app                           Container created
7m35s       Normal   Started                   pod/bar-app                           Container started
7m36s       Normal   SandboxChanged            pod/bar-app                           Pod sandbox changed, it will b
e killed and re-created.
24m         Normal   Sync                      ingress/example-ingress               Scheduled for sync
6m37s       Normal   Sync                      ingress/example-ingress               Scheduled for sync
24m         Normal   Scheduled                 pod/foo-app                           Successfully assigned default/
foo-app to minikube
7m34s       Normal   Pulled                    pod/foo-app                           Container image "docker.io/kic
base/echo-server:1.0" already present on machine and can be accessed by the pod
7m34s       Normal   Created                   pod/foo-app                           Container created
7m34s       Normal   Started                   pod/foo-app                           Container started
7m35s       Normal   SandboxChanged            pod/foo-app                           Pod sandbox changed, it will b
e killed and re-created.
32m         Normal   Scheduled                 pod/hello-minikube-58f7c595dd-77d7p   Successfully assigned default/
hello-minikube-58f7c595dd-77d7p to minikube
32m         Normal   Pulling                   pod/hello-minikube-58f7c595dd-77d7p   Pulling image "kicbase/echo-se
rver:1.0"
32m         Normal   Pulled                    pod/hello-minikube-58f7c595dd-77d7p   Successfully pulled image "kic
base/echo-server:1.0" in 7.127s (7.127s including waiting). Image size: 4939776 bytes.
7m42s       Normal   Created                   pod/hello-minikube-58f7c595dd-77d7p   Container created
7m42s       Normal   Started                   pod/hello-minikube-58f7c595dd-77d7p   Container started
7m45s       Normal   SandboxChanged            pod/hello-minikube-58f7c595dd-77d7p   Pod sandbox changed, it will b
e killed and re-created.
7m43s       Normal   Pulled                    pod/hello-minikube-58f7c595dd-77d7p   Container image "kicbase/echo-
```

```
PS C:\WINDOWS\system32> kubectl config view
apiVersion: v1
clusters:
- cluster:
    certificate-authority: C:\Users\Xue Zhengyang\.minikube\ca.crt
    extensions:
    - extension:
        last-update: Sat, 09 May 2026 18:56:57 CST
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: cluster_info
    server: https://127.0.0.1:59720
  name: minikube
contexts:
- context:
    cluster: minikube
    extensions:
    - extension:
        last-update: Sat, 09 May 2026 18:56:57 CST
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: context_info
    namespace: default
    user: minikube
  name: minikube
current-context: minikube
kind: Config
users:
- name: minikube
  user:
    client-certificate: C:\Users\Xue Zhengyang\.minikube\profiles\minikube\client.crt
    client-key: C:\Users\Xue Zhengyang\.minikube\profiles\minikube\client.key
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl logs hello-node-5f76cf6ccf-br9b5
error: error from server (NotFound): pods "hello-node-5f76cf6ccf-br9b5" not found in namespace "default"
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl get pods
NAME                                READY   STATUS    RESTARTS        AGE
balanced-774d97997-mc8zf            1/1     Running   1 (9m41s ago)   30m
bar-app                             1/1     Running   1 (9m41s ago)   25m
foo-app                             1/1     Running   1 (9m41s ago)   25m
hello-minikube-58f7c595dd-77d7p     1/1     Running   1 (9m40s ago)   34m
hello-node-64fc5894d-2ndhh          1/1     Running   0               3m29s
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl logs hello-node-64fc5894d-2ndhh
I0509 11:03:11.067307       1 log.go:245] Started HTTP server on port 8080
I0509 11:03:11.067688       1 log.go:245] Started UDP server on port  8081
```

```
NOW: 2026-05-09 11:08:23.924788274 +0000 UTC m=+312.890784394
```

```
hello-node-64fc5894d-2ndhh          1/1     Running   0          3m29s
PS C:\WINDOWS\system32> kubectl logs hello-node-64fc5894d-2ndhh
I0509 11:03:11.067307       1 log.go:245] Started HTTP server on port 8080
I0509 11:03:11.067688       1 log.go:245] Started UDP server on port  8081
PS C:\WINDOWS\system32> kubectl expose deployment hello-node --type=LoadBalancer --port=8080
service/hello-node exposed
PS C:\WINDOWS\system32> kubectl get services
NAME             TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
balanced         LoadBalancer   10.99.81.79      127.0.0.1     8080:31750/TCP   32m
bar-service      ClusterIP      10.99.55.81      <none>        8080/TCP         27m
foo-service      ClusterIP      10.104.125.69    <none>        8080/TCP         27m
hello-minikube   NodePort       10.105.184.213   <none>        8080:30858/TCP   36m
hello-node       LoadBalancer   10.111.107.56    <pending>     8080:30544/TCP   12s
kubernetes       ClusterIP      10.96.0.1        <none>        443/TCP          42m
PS C:\WINDOWS\system32> minikube service hello-node
|-----------|------------|-------------|---------------------------|
| NAMESPACE |    NAME    | TARGET PORT |            URL            |
|-----------|------------|-------------|---------------------------|
| default   | hello-node | 8080        | http://192.168.49.2:30544 |
|-----------|------------|-------------|---------------------------|
* 为服务 hello-node 启动隧道。
|-----------|------------|-------------|------------------------|
| NAMESPACE |    NAME    | TARGET PORT |          URL           |
|-----------|------------|-------------|------------------------|
| default   | hello-node |             | http://127.0.0.1:49716 |
|-----------|------------|-------------|------------------------|
* 正通过默认浏览器打开服务 default/hello-node...
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
```

```
* 停止服务 hello-node 的隧道。
PS C:\WINDOWS\system32> minikube addons list
|-----------------------------|----------|--------------|--------------------------------|
|         ADDON NAME          | PROFILE  |    STATUS    |           MAINTAINER           |
|-----------------------------|----------|--------------|--------------------------------|
| ambassador                  | minikube | disabled     | 3rd party (Ambassador)         |
| amd-gpu-device-plugin       | minikube | disabled     | 3rd party (AMD)                |
| auto-pause                  | minikube | disabled     | minikube                       |
| cloud-spanner               | minikube | disabled     | Google                         |
| csi-hostpath-driver         | minikube | disabled     | Kubernetes                     |
| dashboard                   | minikube | enabled ✅   | Kubernetes                     |
| default-storageclass        | minikube | enabled ✅   | Kubernetes                     |
| efk                         | minikube | disabled     | 3rd party (Elastic)            |
| freshpod                    | minikube | disabled     | Google                         |
| gcp-auth                    | minikube | disabled     | Google                         |
| gvisor                      | minikube | disabled     | minikube                       |
| headlamp                    | minikube | disabled     | 3rd party (kinvolk.io)         |
| inaccel                     | minikube | disabled     | 3rd party (InAccel             |
|                             |          |              | [info@inaccel.com])            |
| ingress                     | minikube | enabled ✅   | Kubernetes                     |
| ingress-dns                 | minikube | disabled     | minikube                       |
| inspektor-gadget            | minikube | disabled     | 3rd party                      |
|                             |          |              | (inspektor-gadget.io)          |
| istio                       | minikube | disabled     | 3rd party (Istio)              |
| istio-provisioner           | minikube | disabled     | 3rd party (Istio)              |
| kong                        | minikube | disabled     | 3rd party (Kong HQ)            |
| kubeflow                    | minikube | disabled     | 3rd party                      |
| kubetail                    | minikube | disabled     | 3rd party (kubetail.com)       |
| kubevirt                    | minikube | disabled     | 3rd party (KubeVirt)           |
| logviewer                   | minikube | disabled     | 3rd party (unknown)            |
| metallb                     | minikube | disabled     | 3rd party (MetalLB)            |
| metrics-server              | minikube | disabled     | Kubernetes                     |
| nvidia-device-plugin        | minikube | disabled     | 3rd party (NVIDIA)             |
| nvidia-driver-installer     | minikube | disabled     | 3rd party (NVIDIA)             |
| nvidia-gpu-device-plugin    | minikube | disabled     | 3rd party (NVIDIA)             |
| olm                         | minikube | disabled     | 3rd party (Operator Framework) |
| pod-security-policy         | minikube | disabled     | 3rd party (unknown)            |
| portainer                   | minikube | disabled     | 3rd party (Portainer.io)       |
| registry                    | minikube | disabled     | minikube                       |
| registry-aliases            | minikube | disabled     | 3rd party (unknown)            |
| registry-creds              | minikube | disabled     | 3rd party (UPMC Enterprises)   |
| storage-provisioner         | minikube | enabled ✅   | minikube                       |
| storage-provisioner-rancher | minikube | disabled     | 3rd party (Rancher)            |
| volcano                     | minikube | disabled     | third-party (volcano)          |
| volumesnapshots             | minikube | disabled     | Kubernetes                     |
| yakd                        | minikube | disabled     | 3rd party (marcnuri.com)       |
|-----------------------------|----------|--------------|--------------------------------|
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> minikube addons enable metrics-server
* metrics-server 是由 Kubernetes 维护的插件。如有任何问题，请在 GitHub 上联系 minikube。
您可以在以下链接查看 minikube 的维护者列表：https://github.com/kubernetes/minikube/blob/master/OWNERS
  - 正在使用镜像 registry.k8s.io/metrics-server/metrics-server:v0.8.1
* 启动 'metrics-server' 插件
PS C:\WINDOWS\system32> kubectl get pod,svc -n kube-system
NAME                                   READY   STATUS              RESTARTS      AGE
pod/coredns-7d764666f9-1wrzh           1/1     Running             1 (14m ago)   45m
pod/etcd-minikube                      1/1     Running             1 (14m ago)   45m
pod/kube-apiserver-minikube            1/1     Running             1 (14m ago)   45m
pod/kube-controller-manager-minikube   1/1     Running             1 (14m ago)   45m
pod/kube-proxy-g6z4d                   1/1     Running             1 (14m ago)   45m
pod/kube-scheduler-minikube            1/1     Running             1 (14m ago)   45m
pod/metrics-server-9d74bb658-grxhz     0/1     ContainerCreating   0             10s
pod/storage-provisioner                1/1     Running             2 (14m ago)   45m

NAME                     TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)                  AGE
service/kube-dns         ClusterIP   10.96.0.10     <none>        53/UDP,53/TCP,9153/TCP   45m
service/metrics-server   ClusterIP   10.98.230.186  <none>        443/TCP                  10s
```

```
PS C:\WINDOWS\system32> kubectl top pod
error: Metrics API not available
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl top pod
NAME                               CPU(cores)   MEMORY(bytes)
balanced-774d97997-mc8zf           1m           1Mi
bar-app                            1m           1Mi
foo-app                            1m           1Mi
hello-minikube-58f7c595dd-77d7p    1m           1Mi
hello-node-64fc5894d-2ndhh         1m           6Mi
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl delete service hello-node
>> kubectl delete deployment hello-node
service "hello-node" deleted from default namespace
deployment.apps "hello-node" deleted from default namespace
PS C:\WINDOWS\system32> minikube stop
* 正在停止节点 "minikube" ...
* 正在通过 SSH 关闭 "minikube"…
* 1 个节点已停止。
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl version
Client Version: v1.34.1
Kustomize Version: v5.7.1
Unable to connect to the server: dial tcp 127.0.0.1:8080: connectex: No connection could be made because the target m
achine actively refused it.
PS C:\WINDOWS\system32> kubectl get nodes
E0509 19:15:12.912421   13544 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get
 \"http://localhost:8080/api?timeout=32s\": dial tcp 127.0.0.1:8080: connectex: No connection could be made because t
he target machine actively refused it."
E0509 19:15:12.920336   13544 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get
 \"http://localhost:8080/api?timeout=32s\": dial tcp 127.0.0.1:8080: connectex: No connection could be made because t
he target machine actively refused it."
E0509 19:15:12.921557   13544 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get
 \"http://localhost:8080/api?timeout=32s\": dial tcp 127.0.0.1:8080: connectex: No connection could be made because t
he target machine actively refused it."
E0509 19:15:12.922643   13544 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get
 \"http://localhost:8080/api?timeout=32s\": dial tcp 127.0.0.1:8080: connectex: No connection could be made because t
he target machine actively refused it."
E0509 19:15:12.923666   13544 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get
 \"http://localhost:8080/api?timeout=32s\": dial tcp 127.0.0.1:8080: connectex: No connection could be made because t
he target machine actively refused it."
Unable to connect to the server: dial tcp 127.0.0.1:8080: connectex: No connection could be made because the target m
achine actively refused it.
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl create deployment kubernetes-bootcamp --image=gcr.io/google-samples/kubernetes-bootca
mp:v1
error: failed to create deployment: Post "http://localhost:8080/apis/apps/v1/namespaces/default/deployments?fieldMana
ger=kubectl-create&fieldValidation=Strict": dial tcp 127.0.0.1:8080: connectex: No connection could be made because t
he target machine actively refused it.
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> minikube start
* Microsoft Windows 11 Home China 25H2 上的 minikube v1.38.1
* 根据现有的配置文件使用 docker 驱动程序
* 在集群中 "minikube" 启动节点 "minikube" primary control-plane
* 正在拉取基础镜像 v0.0.50 ...
* 正在为"minikube"重启现有的 docker container ...-
* 正在 Docker 29.2.1 中准备 Kubernetes v1.35.1…\
* 正在验证 Kubernetes 组件...
  - 正在使用镜像 gcr.io/k8s-minikube/storage-provisioner:v5
  - 正在使用镜像 docker.io/kubernetesui/metrics-scraper:v1.0.8
* 插件启用后，请运行 "minikube tunnel" 您的 ingress 资源将在 "127.0.0.1"
  - 正在使用镜像 docker.io/kubernetesui/dashboard:v2.7.0
  - 正在使用镜像 registry.k8s.io/ingress-nginx/controller:v1.14.3
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
  - 正在使用镜像 registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.7
* 正在验证 ingress 插件...
* 某些仪表板功能需要 metrics-server 插件。要启用所有功能，请运行：

        minikube addons enable metrics-server

* 启用插件： storage-provisioner, dashboard, ingress, default-storageclass
* 完成！ kubectl 现在已配置，默认使用"minikube"集群和"default"命名空间
PS C:\WINDOWS\system32> kubectl config current-context
minikube
PS C:\WINDOWS\system32> kubectl create deployment kubernetes-bootcamp --image=gcr.io/google-samples/kubernetes-bootca
mp:v1
deployment.apps/kubernetes-bootcamp created
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
balanced              1/1     1            1           44m
hello-minikube        1/1     1            1           48m
kubernetes-bootcamp   1/1     1            1           2m18s
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS    RESTARTS        AGE
balanced-774d97997-mc8zf               1/1     Running   2 (6m30s ago)   44m
bar-app                                1/1     Running   2 (6m30s ago)   39m
foo-app                                1/1     Running   2 (6m30s ago)   39m
hello-minikube-58f7c595dd-77d7p        1/1     Running   2 (6m30s ago)   48m
kubernetes-bootcamp-67fbdd6b79-scrj2   1/1     Running   0               2m27s
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl proxy
Starting to serve on 127.0.0.1:8001
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> Write-Host "Name of the Pod: $POD_NAME"
Name of the Pod: balanced-774d97997-mc8zf
```

```
PS C:\WINDOWS\system32> Invoke-RestMethod -Uri "http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME`:8080/
proxy/"
Request served by balanced-774d97997-mc8zf

HTTP/1.1 GET /

Host: 127.0.0.1:50921
Accept-Encoding: gzip
User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; zh-CN) WindowsPowerShell/5.1.26100.8115
X-Forwarded-For: 127.0.0.1, 192.168.49.1
X-Forwarded-Uri: /api/v1/namespaces/default/pods/balanced-774d97997-mc8zf:8080/proxy/

PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl delete deployment kubernetes-bootcamp
deployment.apps "kubernetes-bootcamp" deleted from default namespace
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl get pods
NAME                            READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf        1/1     Running   2 (13m ago)   51m
bar-app                         1/1     Running   2 (13m ago)   46m
foo-app                         1/1     Running   2 (13m ago)   46m
hello-minikube-58f7c595dd-77d7p 1/1     Running   2 (13m ago)   55m
PS C:\WINDOWS\system32> kubectl describe pods
Name:             balanced-774d97997-mc8zf
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sat, 09 May 2026 18:35:28 +0800
Labels:           app=balanced
                  pod-template-hash=774d97997
Annotations:      <none>
Status:           Running
IP:               10.244.0.26
IPs:
  IP:           10.244.0.26
Controlled By:  ReplicaSet/balanced-774d97997
Containers:
  echo-server:
    Container ID:   docker://82cdf31ca44b3b0cfcd0e4dd16e9373f73dcd276cc75acf02bb8238980c264da
    Image:          kicbase/echo-server:1.0
    Image ID:       docker-pullable://kicbase/echo-server@sha256:127ac38a2bb9537b7f252addff209ea6801edcac8a92c8b11
acd66a583ed6
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sat, 09 May 2026 19:17:02 +0800
    Last State:     Terminated
      Reason:       Error
      Exit Code:    2
      Started:      Sat, 09 May 2026 18:56:57 +0800
      Finished:     Sat, 09 May 2026 19:13:36 +0800
    Ready:          True
    Restart Count:  2
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-vt2rp (ro)
Conditions:
  Type                          Status
  PodReadyToStartContainers     True
  Initialized                   True
  Ready                         True
  ContainersReady               True
  PodScheduled                  True
Volumes:
  kube-api-access-vt2rp:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason          Age                From               Message
  ----    ------          ----               ----               -------
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> Write-Host "Name of the Pod: $POD_NAME"
>>
>> Invoke-RestMethod -Uri "http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME`:8080/proxy/"
Name of the Pod: balanced-774d97997-mc8zf
Request served by balanced-774d97997-mc8zf

HTTP/1.1 GET /

Host: 127.0.0.1:50921
Accept-Encoding: gzip
User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; zh-CN) WindowsPowerShell/5.1.26100.8115
X-Forwarded-For: 127.0.0.1, 192.168.49.1
X-Forwarded-Uri: /api/v1/namespaces/default/pods/balanced-774d97997-mc8zf:8080/proxy/

PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> Write-Host "Name of the Pod: $POD_NAME"
>>
>> Invoke-RestMethod -Uri "http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME`:8080/proxy/"
Name of the Pod: balanced-774d97997-mc8zf
Request served by balanced-774d97997-mc8zf

HTTP/1.1 GET /

Host: 127.0.0.1:50921
Accept-Encoding: gzip
User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; zh-CN) WindowsPowerShell/5.1.26100.8115
X-Forwarded-For: 127.0.0.1, 192.168.49.1
X-Forwarded-Uri: /api/v1/namespaces/default/pods/balanced-774d97997-mc8zf:8080/proxy/

PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> kubectl exec $POD_NAME -- env
OCI runtime exec failed: exec failed: unable to start container process: exec: "env": executable file not found in $P
ATH
command terminated with exit code 127
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> kubectl exec -ti $POD_NAME -- bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "bash": executable file not found in $
PATH
command terminated with exit code 127
PS C:\WINDOWS\system32> _
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> kubectl exec -ti $POD_NAME -- bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "bash": executable file not found in $
PATH
command terminated with exit code 127
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> kubectl exec -ti $POD_NAME -- sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "sh": executable file not found in $PA
TH
command terminated with exit code 127
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> kubectl exec -ti $POD_NAME -- sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "sh": executable file not found in $PA
TH
command terminated with exit code 127
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> kubectl logs $POD_NAME
Echo server listening on port 8080.
10.244.0.1:47492 | GET /
10.244.0.1:39278 | GET /
PS C:\WINDOWS\system32>
```

```
Request served by balanced-774d97997-mc8zf

HTTP/1.1 GET /

Host: localhost:8080
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-
exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate, br, zstd
Accept-Language: zh-CN,zh;q=0.9,en;q=0.8,en-GB;q=0.7,en-US;q=0.6
Connection: keep-alive
Cookie: Pycharm-395552fc=ec6381ac-3911-4a39-a14d-469ea85efc5b; Hm_lvt_a1ff8825baa73c3a78eb96aa40325abc=1755144592; Clion-
4e73588f=88c5b04e-1da8-46d5-98aa-67711f5951c9;
gitea_incredible=4yUfCzm2eI%3Aebfa4eb772aaa4d3a30469b5fadbc7bf6c1404e22db2763ca2c0c6571182ae50
Sec-Ch-Ua: "Chromium";v="148", "Microsoft Edge";v="148", "Not/A)Brand";v="99"
Sec-Ch-Ua-Mobile: ?0
Sec-Ch-Ua-Platform: "Windows"
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: none
Sec-Fetch-User: ?1
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
Edg/148.0.0.0
```

```
PS C:\WINDOWS\system32> kubectl get pods
NAME                         READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf     1/1     Running   2 (23m ago)   61m
bar-app                      1/1     Running   2 (23m ago)   56m
foo-app                      1/1     Running   2 (23m ago)   56m
hello-minikube-58f7c595dd-77d7p   1/1   Running   2 (23m ago)   65m
PS C:\WINDOWS\system32> kubectl create deployment kubernetes-bootcamp --image=gcr.io/google-samples/kubernetes-bootca
mp:v1
deployment.apps/kubernetes-bootcamp created
```

```
PS C:\WINDOWS\system32> kubectl get services
NAME             TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
balanced         LoadBalancer   10.99.81.79      127.0.0.1     8080:31750/TCP   62m
bar-service      ClusterIP      10.99.55.81      <none>        8080/TCP         57m
foo-service      ClusterIP      10.104.125.69    <none>        8080/TCP         57m
hello-minikube   NodePort       10.105.184.213   <none>        8080:30858/TCP   65m
kubernetes       ClusterIP      10.96.0.1        <none>        443/TCP          71m
PS C:\WINDOWS\system32> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
service/kubernetes-bootcamp exposed
PS C:\WINDOWS\system32> kubectl describe services/kubernetes-bootcamp
Name:                     kubernetes-bootcamp
Namespace:                default
Labels:                   app=kubernetes-bootcamp
Annotations:              <none>
Selector:                 app=kubernetes-bootcamp
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.108.37.204
IPs:                      10.108.37.204
Port:                     <unset>  8080/TCP
TargetPort:               8080/TCP
NodePort:                 <unset>  32537/TCP
Endpoints:                10.244.0.29:8080
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
PS C:\WINDOWS\system32> _
```

```
PS C:\WINDOWS\system32> $NODE_PORT = (kubectl get services/kubernetes-bootcamp -o jsonpath='{.spec.ports[0].nodePort}'
)
>> Write-Host "NODE_PORT=$NODE_PORT"
NODE_PORT=32537
PS C:\WINDOWS\system32>
```

```
NAME                                   READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf               1/1     Running   2 (23m ago)   61m
bar-app                                1/1     Running   2 (23m ago)   56m
foo-app                                1/1     Running   2 (23m ago)   56m
hello-minikube-58f7c595dd-77d7p        1/1     Running   2 (23m ago)   65m
kubernetes-bootcamp-67fbdd6b79-b2vns   1/1     Running   0             9s
PS C:\WINDOWS\system32> kubectl get services
NAME             TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
balanced         LoadBalancer   10.99.81.79      127.0.0.1     8080:31750/TCP   62m
bar-service      ClusterIP      10.99.55.81      <none>        8080/TCP         57m
foo-service      ClusterIP      10.104.125.69    <none>        8080/TCP         57m
hello-minikube   NodePort       10.105.184.213   <none>        8080:30858/TCP   65m
kubernetes       ClusterIP      10.96.0.1        <none>        443/TCP          71m
PS C:\WINDOWS\system32> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
service/kubernetes-bootcamp exposed
PS C:\WINDOWS\system32> kubectl describe services/kubernetes-bootcamp
Name:                     kubernetes-bootcamp
Namespace:                default
Labels:                   app=kubernetes-bootcamp
Annotations:              <none>
Selector:                 app=kubernetes-bootcamp
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.108.37.204
IPs:                      10.108.37.204
Port:                     <unset>  8080/TCP
TargetPort:               8080/TCP
NodePort:                 <unset>  32537/TCP
Endpoints:                10.244.0.29:8080
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
PS C:\WINDOWS\system32> $NODE_PORT = (kubectl get services/kubernetes-bootcamp -o jsonpath='{.spec.ports[0].nodePort}
')
>> Write-Host "NODE_PORT=$NODE_PORT"
NODE_PORT=32537
PS C:\WINDOWS\system32> minikube service kubernetes-bootcamp --url
http://127.0.0.1:59578
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
```

```
PS C:\WINDOWS\system32> curl http://127.0.0.1:59578

安全警告: 脚本执行风险
Invoke-WebRequest 可解析网页内容。解析网页时，可能会运行网页中的脚本代码。
      建议的操作:
      使用 -UseBasicParsing 开关来避免执行脚本代码。

      是否要继续?

[Y] 是(Y)  [A] 全是(A)  [N] 否(N)  [L] 全否(L)  [S] 暂停(S)  [?] 帮助 (默认值为“N”): Y


StatusCode        : 200
StatusDescription : OK
Content           : Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-b2vns | v=1

RawContent        : HTTP/1.1 200 OK
                    Connection: keep-alive
                    Transfer-Encoding: chunked
                    Content-Type: text/plain
                    Date: Sat, 09 May 2026 11:40:15 GMT

                    Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbd...
Forms             : {}
Headers           : {[Connection, keep-alive], [Transfer-Encoding, chunked], [Content-Type, text/plain], [Date, Sat, 09
                     May 2026 11:40:15 GMT]}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 84



PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl describe deployment
Name:                   balanced
Namespace:              default
CreationTimestamp:      Sat, 09 May 2026 18:35:28 +0800
Labels:                 app=balanced
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=balanced
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=balanced
  Containers:
   echo-server:
    Image:         kicbase/echo-server:1.0
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets:  <none>
NewReplicaSet:   balanced-774d97997 (1/1 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  66m   deployment-controller  Scaled up replica set balanced-774d97997 from 0 to 1


Name:                   hello-minikube
Namespace:              default
CreationTimestamp:      Sat, 09 May 2026 18:31:41 +0800
Labels:                 app=hello-minikube
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=hello-minikube
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=hello-minikube
  Containers:
   echo-server:
    Image:         kicbase/echo-server:1.0
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
```

```
PS C:\WINDOWS\system32> kubectl get pods -l app=kubernetes-bootcamp
>> kubectl get services -l app=kubernetes-bootcamp
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-67fbdd6b79-b2vns   1/1     Running   0          5m20s
NAME                  TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes-bootcamp   NodePort   10.108.37.204   <none>        8080:32537/TCP   4m44s
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath='{.items[0].metadata.name}')
>> kubectl label pods $POD_NAME version=v1
>>
pod/balanced-774d97997-mc8zf labeled
PS C:\WINDOWS\system32> kubectl describe pods $POD_NAME
>> kubectl get pods -l version=v1
Name:             balanced-774d97997-mc8zf
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sat, 09 May 2026 18:35:28 +0800
Labels:           app=balanced
                  pod-template-hash=774d97997
                  version=v1
Annotations:      <none>
Status:           Running
IP:               10.244.0.26
IPs:
  IP:           10.244.0.26
Controlled By:  ReplicaSet/balanced-774d97997
Containers:
  echo-server:
    Container ID:   docker://82cdf31ca44b3b0cfcd0e4dd16e9373f73dcd276cc75acf02bb8238980c264da
    Image:          kicbase/echo-server:1.0
    Image ID:       docker-pullable://kicbase/echo-server@sha256:127ac38a2bb9537b7f252addff209ea6801edcac8a92c8b1104
acd66a583ed6
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sat, 09 May 2026 19:17:02 +0800
    Last State:     Terminated
      Reason:       Error
      Exit Code:    2
```

```
PS C:\WINDOWS\system32> kubectl delete service -l app=kubernetes-bootcamp
>> kubectl get services
service "kubernetes-bootcamp" deleted from default namespace
NAME             TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
balanced         LoadBalancer   10.99.81.79      127.0.0.1     8080:31750/TCP   67m
bar-service      ClusterIP      10.99.55.81      <none>        8080/TCP         62m
foo-service      ClusterIP      10.104.125.69    <none>        8080/TCP         62m
hello-minikube   NodePort       10.105.184.213   <none>        8080:30858/TCP   71m
kubernetes       ClusterIP      10.96.0.1        <none>        443/TCP          77m
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
balanced              1/1     1            1           103m
hello-minikube        1/1     1            1           107m
kubernetes-bootcamp   1/1     1            1           41m
PS C:\WINDOWS\system32> kubectl get rs
NAME                             DESIRED   CURRENT   READY   AGE
balanced-774d97997               1         1         1       103m
hello-minikube-58f7c595dd        1         1         1       107m
kubernetes-bootcamp-67fbdd6b79   1         1         1       42m
PS C:\WINDOWS\system32> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
service/kubernetes-bootcamp exposed
PS C:\WINDOWS\system32> kubectl scale deployments/kubernetes-bootcamp --replicas=4
deployment.apps/kubernetes-bootcamp scaled
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
balanced              1/1     1            1           105m
hello-minikube        1/1     1            1           108m
kubernetes-bootcamp   4/4     4            4           43m
PS C:\WINDOWS\system32> kubectl get pods -o wide
NAME                                   READY   STATUS    RESTARTS      AGE    IP            NODE       NOMINATED NODE
   READINESS GATES
balanced-774d97997-mc8zf               1/1     Running   2 (67m ago)   105m   10.244.0.26   minikube   <none>
   <none>
bar-app                                1/1     Running   2 (67m ago)   100m   10.244.0.25   minikube   <none>
   <none>
foo-app                                1/1     Running   2 (67m ago)   100m   10.244.0.21   minikube   <none>
   <none>
hello-minikube-58f7c595dd-77d7p        1/1     Running   2 (67m ago)   108m   10.244.0.24   minikube   <none>
   <none>
kubernetes-bootcamp-67fbdd6b79-4jf6t   1/1     Running   0             18s    10.244.0.31   minikube   <none>
   <none>
kubernetes-bootcamp-67fbdd6b79-b2vns   1/1     Running   0             43m    10.244.0.29   minikube   <none>
   <none>
kubernetes-bootcamp-67fbdd6b79-qd878   1/1     Running   0             18s    10.244.0.30   minikube   <none>
   <none>
kubernetes-bootcamp-67fbdd6b79-wrlh2   1/1     Running   0             18s    10.244.0.32   minikube   <none>
   <none>
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl describe deployments/kubernetes-bootcamp
Name:                   kubernetes-bootcamp
Namespace:              default
CreationTimestamp:      Sat, 09 May 2026 19:37:09 +0800
Labels:                 app=kubernetes-bootcamp
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=kubernetes-bootcamp
Replicas:               4 desired | 4 updated | 4 total | 4 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
   kubernetes-bootcamp:
    Image:        gcr.io/google-samples/kubernetes-bootcamp:v1
    Port:         <none>
    Host Port:    <none>
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
  Node-Selectors: <none>
  Tolerations:    <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets:  <none>
NewReplicaSet:   kubernetes-bootcamp-67fbdd6b79 (4/4 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  44m   deployment-controller  Scaled up replica set kubernetes-bootcamp-67fbdd6b79 from 0
 to 1
  Normal  ScalingReplicaSet  82s   deployment-controller  Scaled up replica set kubernetes-bootcamp-67fbdd6b79 from 1
 to 4
PS C:\WINDOWS\system32> _
```

```
版权所有（C） Microsoft Corporation。保留所有权利。
                                                        安装最新的 PowerShell，了解新功能和改进！https://aka.ms/PSWindows
                                                          PS C:\WINDOWS\system32> curl http://127.0.0.1:58393

安全警告: 脚本执行风险
Invoke-WebRequest 可解析网页内容。解析网页时，可能会运行网页中的脚本代码。
      建议的操作:
      使用 -UseBasicParsing 开关来避免执行脚本代码。

      是否要继续?
[Y] 是(Y)  [A] 全是(A)  [N] 否(N)  [L] 全否(L)  [S] 暂停(S)  [?] 帮助 (默认值为“N”): Y


StatusCode        : 200
StatusDescription : OK
Content           : Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-b2vns | v=1

RawContent        : HTTP/1.1 200 OK
                    Connection: keep-alive
                    Transfer-Encoding: chunked
                    Content-Type: text/plain
                    Date: Sat, 09 May 2026 12:22:56 GMT

                    Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbd...
Forms             : {}
Headers           : {[Connection, keep-alive], [Transfer-Encoding, chunked], [Content-Type, text/plain], [Date, Sat, 09
                     May 2026 12:22:56 GMT]}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 84



PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl scale deployments/kubernetes-bootcamp --replicas=2
deployment.apps/kubernetes-bootcamp scaled
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl scale deployments/kubernetes-bootcamp --replicas=2
deployment.apps/kubernetes-bootcamp scaled
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
balanced              1/1     1            1           110m
hello-minikube        1/1     1            1           113m
kubernetes-bootcamp   2/2     2            2           48m
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf               1/1     Running   2 (72m ago)   110m
bar-app                                1/1     Running   2 (72m ago)   105m
foo-app                                1/1     Running   2 (72m ago)   105m
hello-minikube-58f7c595dd-77d7p        1/1     Running   2 (72m ago)   114m
kubernetes-bootcamp-67fbdd6b79-4jf6t   1/1     Running   0             5m23s
kubernetes-bootcamp-67fbdd6b79-b2vns   1/1     Running   0             48m
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl get pods
NAME                                  READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf              1/1     Running   2 (72m ago)   110m
bar-app                               1/1     Running   2 (72m ago)   105m
foo-app                               1/1     Running   2 (72m ago)   105m
hello-minikube-58f7c595dd-77d7p       1/1     Running   2 (72m ago)   114m
kubernetes-bootcamp-67fbdd6b79-4jf6t  1/1     Running   0             5m23s
kubernetes-bootcamp-67fbdd6b79-b2vns  1/1     Running   0             48m
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
balanced              1/1     1            1           112m
hello-minikube        1/1     1            1           116m
kubernetes-bootcamp   2/2     2            2           50m
PS C:\WINDOWS\system32> kubectl get pods
NAME                                  READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf              1/1     Running   2 (74m ago)   112m
bar-app                               1/1     Running   2 (74m ago)   107m
foo-app                               1/1     Running   2 (74m ago)   107m
hello-minikube-58f7c595dd-77d7p       1/1     Running   2 (74m ago)   116m
kubernetes-bootcamp-67fbdd6b79-4jf6t  1/1     Running   0             7m36s
kubernetes-bootcamp-67fbdd6b79-b2vns  1/1     Running   0             50m
PS C:\WINDOWS\system32> kubectl describe pods | Select-String "Image:"

    Image:          kicbase/echo-server:1.0
    Image:          docker.io/kicbase/echo-server:1.0
    Image:          docker.io/kicbase/echo-server:1.0
    Image:          kicbase/echo-server:1.0
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v1
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v1


PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=docker.io/jocatalin/kub
ernetes-bootcamp:v2
deployment.apps/kubernetes-bootcamp image updated
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> kubectl rollout status deployments/kubernetes-bootcamp
deployment "kubernetes-bootcamp" successfully rolled out
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=docker.io/jocatalin/kub
ernetes-bootcamp:v2
deployment.apps/kubernetes-bootcamp image updated
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> kubectl rollout status deployments/kubernetes-bootcamp
deployment "kubernetes-bootcamp" successfully rolled out
PS C:\WINDOWS\system32> kubectl get pods -w
NAME                                  READY   STATUS        RESTARTS      AGE
balanced-774d97997-mc8zf              1/1     Running       2 (75m ago)   113m
bar-app                               1/1     Running       2 (75m ago)   108m
foo-app                               1/1     Running       2 (75m ago)   108m
hello-minikube-58f7c595dd-77d7p       1/1     Running       2 (75m ago)   117m
kubernetes-bootcamp-67fbdd6b79-4jf6t  1/1     Terminating   0             8m51s
kubernetes-bootcamp-67fbdd6b79-b2vns  1/1     Terminating   0             52m
kubernetes-bootcamp-68cf47c9c8-pk4nm  1/1     Running       0             38s
kubernetes-bootcamp-68cf47c9c8-zs9kc  1/1     Running       0             29s
kubernetes-bootcamp-67fbdd6b79-4jf6t  0/1     Error         0             8m52s
kubernetes-bootcamp-67fbdd6b79-4jf6t  0/1     Error         0             8m53s
kubernetes-bootcamp-67fbdd6b79-4jf6t  0/1     Error         0             8m53s
kubernetes-bootcamp-67fbdd6b79-4jf6t  0/1     Error         0             8m53s
kubernetes-bootcamp-67fbdd6b79-b2vns  0/1     Error         0             52m
kubernetes-bootcamp-67fbdd6b79-b2vns  0/1     Error         0             52m
kubernetes-bootcamp-67fbdd6b79-b2vns  0/1     Error         0             52m
kubernetes-bootcamp-67fbdd6b79-b2vns  0/1     Error         0             52m
```

```
PS C:\WINDOWS\system32> kubectl get pods -w
NAME                                  READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf              1/1     Running   2 (76m ago)   114m
bar-app                               1/1     Running   2 (76m ago)   109m
foo-app                               1/1     Running   2 (76m ago)   109m
hello-minikube-58f7c595dd-77d7p       1/1     Running   2 (76m ago)   118m
kubernetes-bootcamp-68cf47c9c8-pk4nm  1/1     Running   0             93s
kubernetes-bootcamp-68cf47c9c8-zs9kc  1/1     Running   0             84s
```

```
PS C:\WINDOWS\system32> kubectl describe pods | Select-String "Image:"

    Image:          kicbase/echo-server:1.0
    Image:          docker.io/kicbase/echo-server:1.0
    Image:          docker.io/kicbase/echo-server:1.0
    Image:          kicbase/echo-server:1.0
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2


PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
Error from server (AlreadyExists): services "kubernetes-bootcamp" already exists
PS C:\WINDOWS\system32> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=gcr.io/google-samples/k
ubernetes-bootcamp:v10
deployment.apps/kubernetes-bootcamp image updated
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
balanced              1/1     1            1           116m
hello-minikube        1/1     1            1           119m
kubernetes-bootcamp   2/2     1            2           54m
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS             RESTARTS      AGE
balanced-774d97997-mc8zf               1/1     Running            2 (78m ago)   116m
bar-app                                1/1     Running            2 (78m ago)   111m
foo-app                                1/1     Running            2 (78m ago)   111m
hello-minikube-58f7c595dd-77d7p        1/1     Running            2 (78m ago)   119m
kubernetes-bootcamp-68cf47c9c8-pk4nm   1/1     Running            0             3m8s
kubernetes-bootcamp-68cf47c9c8-zs9kc   1/1     Running            0             2m59s
kubernetes-bootcamp-845bc89d75-xlwvj   0/1     ImagePullBackOff   0             30s
PS C:\WINDOWS\system32> kubectl describe pods
Name:             balanced-774d97997-mc8zf
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sat, 09 May 2026 18:35:28 +0800
Labels:           app=balanced
                  pod-template-hash=774d97997
                  version=v1
Annotations:      <none>
Status:           Running
IP:               10.244.0.26
IPs:
  IP:           10.244.0.26
Controlled By:  ReplicaSet/balanced-774d97997
Containers:
  echo-server:
    Container ID:   docker://82cdf31ca44b3b0cfcd0e4dd16e9373f73dcd276cc75acf02bb8238980c264da
    Image:          kicbase/echo-server:1.0
    Image ID:       docker-pullable://kicbase/echo-server@sha256:127ac38a2bb9537b7f252addff209ea6801edcac8a92c8b1104d
acd66a583ed6
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Sat, 09 May 2026 19:17:02 +0800
    Last State:     Terminated
      Reason:       Error
      Exit Code:    2
      Started:      Sat, 09 May 2026 18:56:57 +0800
      Finished:     Sat, 09 May 2026 19:13:36 +0800
    Ready:          True
    Restart Count:  2
```

```
Events:
  Type     Reason     Age                From               Message
  ----     ------     ----               ----               -------
  Normal   Scheduled  38s                default-scheduler  Successfully assigned default/kubernetes-bootcamp-845bc89
d75-xlwvj to minikube
  Normal   Pulling    18s (x2 over 37s)  kubelet            Pulling image "gcr.io/google-samples/kubernetes-bootcamp:
v10"
  Warning  Failed     12s (x2 over 30s)  kubelet            Failed to pull image "gcr.io/google-samples/kubernetes-bo
otcamp:v10": Error response from daemon: manifest for gcr.io/google-samples/kubernetes-bootcamp:v10 not found: manife
st unknown: Failed to fetch "v10"
  Warning  Failed     12s (x2 over 30s)  kubelet            Error: ErrImagePull
  Normal   BackOff    0s (x2 over 30s)   kubelet            Back-off pulling image "gcr.io/google-samples/kubernetes-
bootcamp:v10"
  Warning  Failed     0s (x2 over 30s)   kubelet            Error: ImagePullBackOff
PS C:\WINDOWS\system32>
```

```
  Warning  Failed     6s (x2 over 30s)   kubelet            Error: ImagePullBackOff
PS C:\WINDOWS\system32> kubectl rollout undo deployments/kubernetes-bootcamp
deployment.apps/kubernetes-bootcamp rolled back
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS    RESTARTS      AGE
balanced-774d97997-mc8zf               1/1     Running   2 (78m ago)   117m
bar-app                                1/1     Running   2 (78m ago)   112m
foo-app                                1/1     Running   2 (78m ago)   112m
hello-minikube-58f7c595dd-77d7p        1/1     Running   2 (78m ago)   120m
kubernetes-bootcamp-68cf47c9c8-pk4nm   1/1     Running   0             4m
kubernetes-bootcamp-68cf47c9c8-zs9kc   1/1     Running   0             3m51s
PS C:\WINDOWS\system32> kubectl describe pods | Select-String "Image:"

    Image:          kicbase/echo-server:1.0
    Image:          docker.io/kicbase/echo-server:1.0
    Image:          docker.io/kicbase/echo-server:1.0
    Image:          kicbase/echo-server:1.0
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2
    Image:          docker.io/jocatalin/kubernetes-bootcamp:v2


PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl delete deployments/kubernetes-bootcamp services/kubernetes-bootcamp
deployment.apps "kubernetes-bootcamp" deleted from default namespace
service "kubernetes-bootcamp" deleted from default namespace
PS C:\WINDOWS\system32>
```